

Homework BE1: Simulation of a tuning fork

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During this BE session, we will consider the differential equation that models a tuning fork, that is:

$$\frac{d^2x(t)}{dt^2} + \frac{f}{m} \frac{dx(t)}{dt} + \frac{k}{m}x(t) = E(t) \quad (1)$$

$$\begin{cases} x(0) = x_0 \neq 0 \\ x'(0) = 0 \end{cases} \quad (2)$$

where $x(t)$ is the deviation of the tuning fork branch, m the mass, f the rigidity coefficient, k the friction coefficient, $E(t)$ the input and x_0 a constant different than 0.

We will consider two situations:

1. $E(t) = 0$.
2. $E(t) = A \sin(\omega_E t) \varepsilon(t)$.

Let us notice that $\frac{f}{m} = f_0 = 2\alpha$ and $\frac{k}{m} = k_0 = \alpha^2 + \omega_0^2$:

$$\frac{d^2x(t)}{dt^2} + f_0 \frac{dx(t)}{dt} + k_0 x(t) = E(t) \quad (3)$$

$$\frac{d^2x(t)}{dt^2} + 2\alpha \frac{dx(t)}{dt} + (\alpha^2 + \omega_0^2)x(t) = E(t) \quad (4)$$

First situation: $E(t) = 0$

1. Using the formulation of α and ω , solve the characteristic equation and find the solution of this homogeneous equation satisfying the initial conditions (i.e., find the values of c_1 and c_2 using the initial conditions). How do you interpret the coefficients α and ω_0 ?

For $E(t) = 0$, the differential equation is:

$$\frac{d^2x}{dt^2} + 2\alpha \frac{dx}{dt} + (\alpha^2 + \omega_0^2)x = 0$$

Assuming a solution of the form $x(t) = e^{rt}$, the characteristic equation is:

$$r^2 + 2\alpha r + (\alpha^2 + \omega_0^2) = 0$$

Using the quadratic formula, the roots are given by:

$$r_{1,2} = \frac{-2\alpha \pm \sqrt{(2\alpha)^2 - 4(\alpha^2 + \omega_0^2)}}{2}$$

Simplifying the discriminant:

$$\sqrt{4\alpha^2 - 4(\alpha^2 + \omega_0^2)} = \sqrt{4\alpha^2 - 4\alpha^2 - 4\omega_0^2} = \sqrt{-4\omega_0^2} = 2i\omega_0$$

Thus, the roots of the characteristic equation are:

$$r_{1,2} = \frac{-2\alpha \pm 2i\omega_0}{2} = -\alpha \pm i\omega_0$$

Since the roots are complex conjugates, the general solution to the differential equation is:

$$x(t) = e^{-\alpha t} (c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t))$$

To determine the constants c_1 and c_2 , we apply the initial conditions $x(0) = x_0$ and $x'(0) = 0$.

At $t = 0$, the solution becomes:

$$x(0) = e^0 (c_1 \cos(0) + c_2 \sin(0)) = c_1 \cdot 1 + c_2 \cdot 0 = c_1$$

Therefore, $c_1 = x_0$.

The derivative of $x(t)$ is:

$$x'(t) = -\alpha e^{-\alpha t} (c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t)) + e^{-\alpha t} (-c_1 \omega_0 \sin(\omega_0 t) + c_2 \omega_0 \cos(\omega_0 t))$$

At $t = 0$, the derivative simplifies to:

$$x'(0) = -\alpha e^0 (c_1 \cos(0) + c_2 \sin(0)) + e^0 (-c_1 \omega_0 \sin(0) + c_2 \omega_0 \cos(0))$$

$$x'(0) = -\alpha(c_1 \cdot 1 + c_2 \cdot 0) + (-c_1 \omega_0 \cdot 0 + c_2 \omega_0 \cdot 1)$$

$$x'(0) = -\alpha c_1 + c_2 \omega_0$$

Since $x'(0) = 0$, we have:

$$-\alpha c_1 + c_2 \omega_0 = 0$$

Substituting $c_1 = x_0$, we solve for c_2 :

$$c_2 = \frac{\alpha x_0}{\omega_0}$$

Substituting c_1 and c_2 into the general solution, we get:

$$x(t) = e^{-\alpha t} \left(x_0 \cos(\omega_0 t) + \frac{\alpha x_0}{\omega_0} \sin(\omega_0 t) \right)$$

Interpretation of coefficients α and ω_0

- **damping factor (α):** damping of the system occurs due to internal friction and air resistance. A higher value of α means that the oscillations of the tuning fork will end more quickly. This results in the sound of the tuning fork fading faster over time.
- **characteristic frequency (ω_0):** the characteristic frequency ω_0 is the natural angular frequency of the tuning fork, determined by its material properties and physical shape. It defines how fast the tuning fork vibrates when struck and is directly related to the pitch of the sound it produces. Specifically, $\omega_0 = 2\pi F_0$, where F_0 is the frequency of the tuning fork in Hz.

2. Using the formulation with f_0 and k_0 , find a block diagram with adders, multipliers, and integrators to simulate this differential equation. Define the initial condition of each integrator.

The block diagram represents the differential equation:

$$\frac{d^2x}{dt^2} + 2\alpha\frac{dx}{dt} + (\alpha^2 + \omega_0^2)x = 0$$

The diagram consists of the following components:

- **integrators:**
 - $u1$: computes $\frac{dx}{dt}$ from $\frac{d^2x}{dt^2}$.
 - $u2$: computes $x(t)$ from $\frac{dx}{dt}$.

The initial conditions of the integrators are:

- $u_1(0) = x'(0) + 2\alpha x(0) = 2\alpha x_0$ (since $x'(0) = 0$).
- $u_2(0) = x(0) = x_0$.
- **adders:** combine terms to form the input to the first integrator.
- **gain blocks:**
 - $gain1$: scales the signal by $f_0 = 2\alpha$.
 - $gain2$: scales the signal by $k_0 = \alpha^2 + \omega_0^2$.

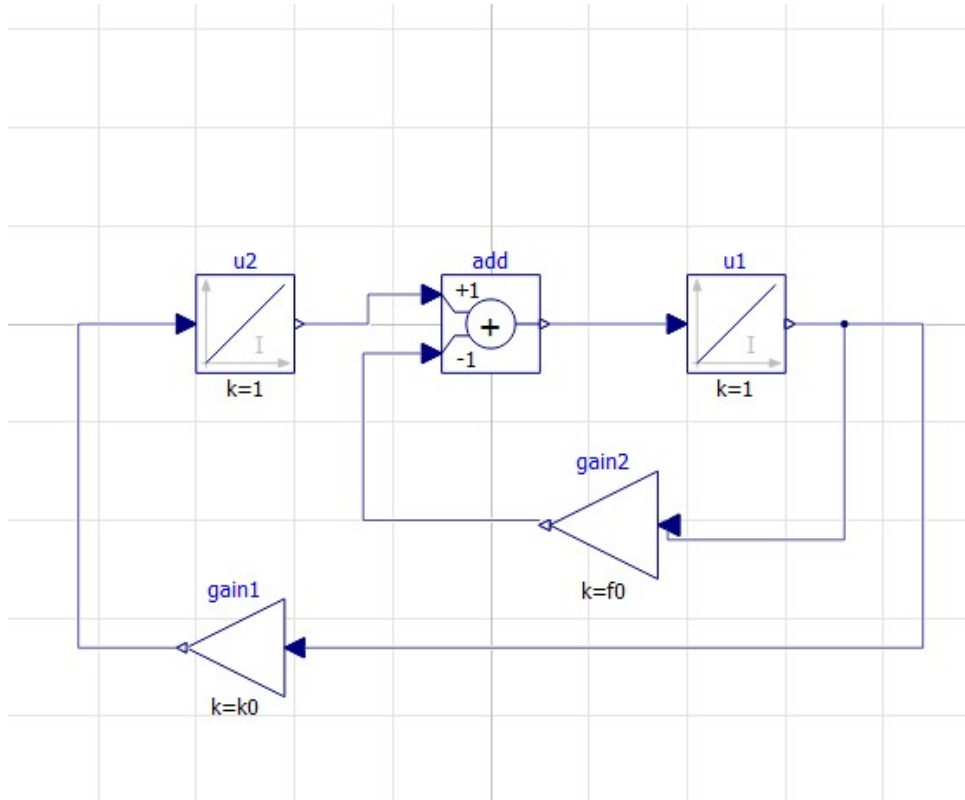


Figure 1: Block diagram in OpenModelica

The OpenModelica code used for the block diagram is:

```

import Modelica.Constants.pi;
parameter Real x0=5;
parameter Real F0=440;
parameter Real alpha=F0/10;
parameter Real k0=alpha^2+(2*pi*F0)^2;
parameter Real f0=2*alpha;

Modelica.Blocks.Continuous.Integrator u1;
Modelica.Blocks.Continuous.Integrator u2;
Modelica.Blocks.Math.Gain gain1(k = f0);
Modelica.Blocks.Math.Add add(k2 = -1);
Modelica.Blocks.Math.Gain gain2(k = k0);

equation
connect(u1.y, add.u1);
connect(add.y, u2.u);
connect(gain1.y, add.u2);
connect(u1.u, gain2.u);
connect(u2.y, gain2.u);
connect(u2.y, gain1.u);
annotation(
uses(Modelica(version = "4.0.0")));
end diapason;

```

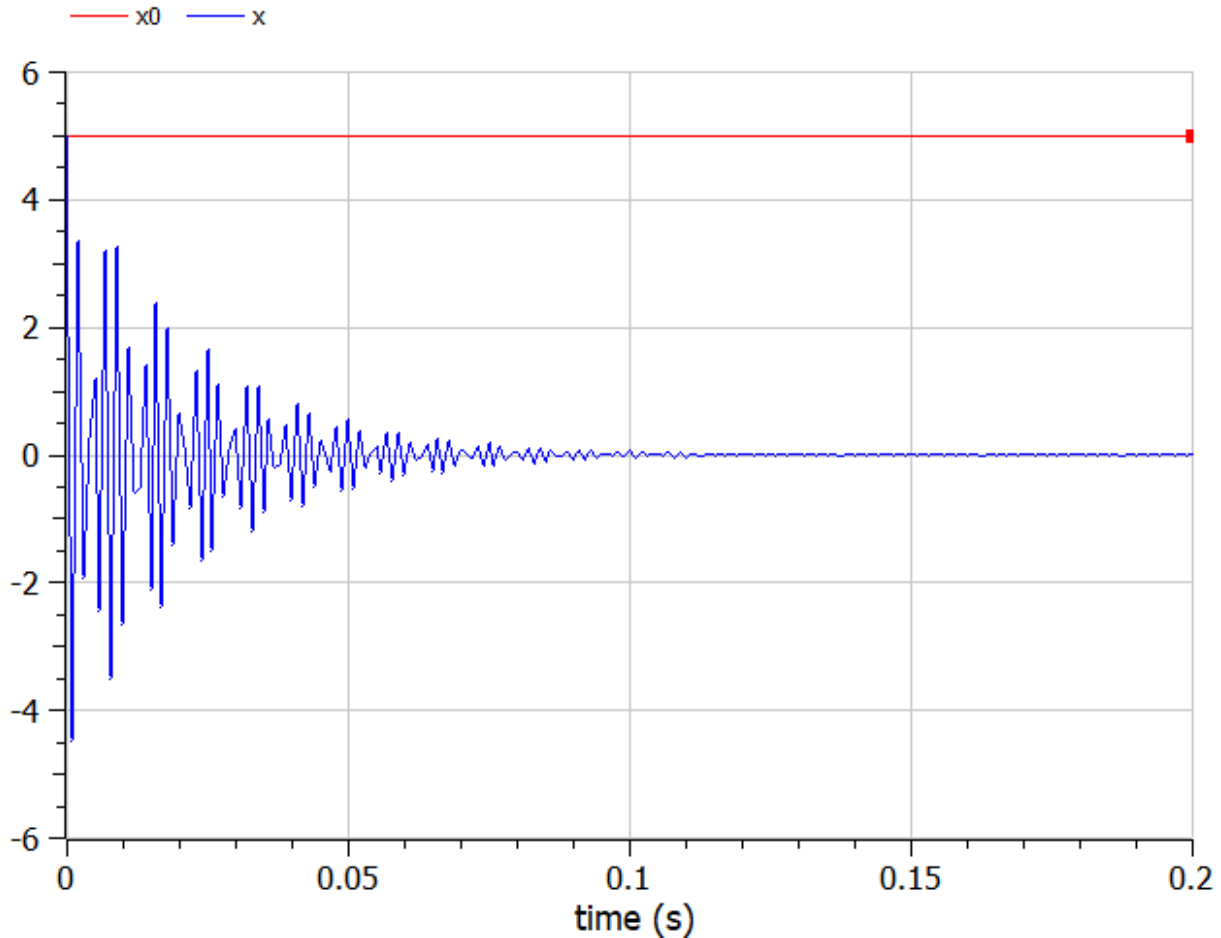


Figure 2: Simulation of 200ms in OpenModelica

3. Implement this dynamic system in OpenModelica (either by using equations or the block diagram). Use the following values: $x_0 = 5$, $\alpha = \frac{F_0}{10}$ and $F_0 = 440$ Hz (take into account that $\omega_0 = 2\pi F_0$). Run a simulation of 200ms. Verify your simulation with the solution found in Q1.

- Are the initial conditions correct?
- What can you say about the characteristic frequency?
- And the damping factor?

The code used in OpenModelica for the implementation is:

```
model TFFree_eq

import Modelica.Constants.pi;
parameter Real x0=5;
parameter Real F0=440;
parameter Real alpha=F0/10;
parameter Real k0=alpha^2+(2*pi*F0)^2;
parameter Real f0=2*alpha;
```

```

Real u1(start =f0);
Real u2(start =x0);
Real x;

equation
x=u2;
der(u1)=-k0*x;
der(u2)=-f0*x+u1;

annotation(
  experiment(
    StartTime=0,
    StopTime=0.2,
    Interval=0.001
  ),
  uses(Modelica(version = "4.0.0"))
);

end TFFree_eq;

```

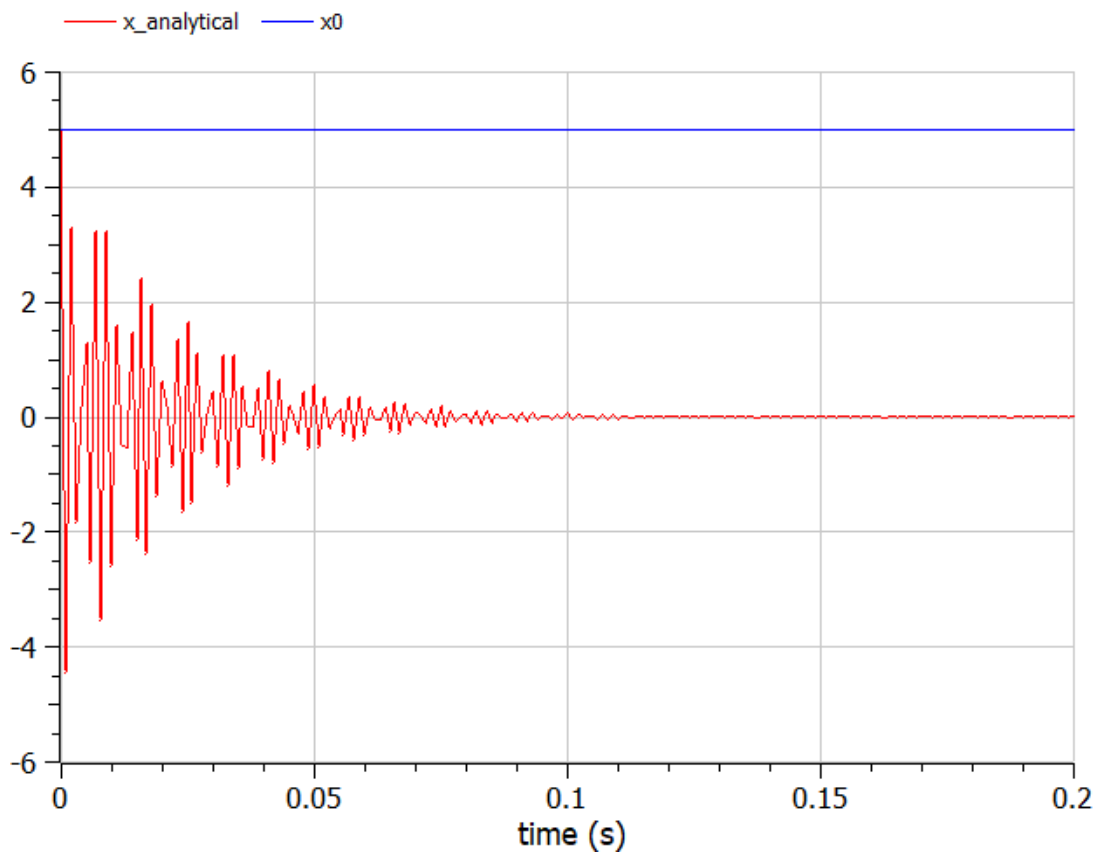


Figure 3: Analytical solution in OpenModelica

- The simulation results closely match the analytical solution derived in Question 1.
- Minor discrepancies are due to numerical approximations in the simulation.

- The initial conditions are $x(0) = 5$, $u_2(0) = x_0 = 5$ and $u_1(0) = x'(0) + 2\alpha \times x(0) = 0 + 2 \times 44 \times 5 = 440$.
- The simulation shows that indeed $x(0) = 5$ and $u_2(0) = x_0 = 5$, but we can see that $u_1(0) = 88$, so they are not fully correct.
- The characteristic frequency has the value $\omega_0 = 2764.60$, which is perfectly equal to $2 \times \pi \times F_0 = 2 \times \pi \times 440 = 2764.60$ Hz.
- The damping factor is $\alpha = 44$, the exact same value as $\frac{F_0}{10} = \frac{440}{10} = 44s^{-1}$.

The code used in OpenModelica for the implementation of the analytical solution is:

```
model solution

import Modelica.Constants.pi;
parameter Real x0=5;
parameter Real F0=440;
parameter Real alpha=F0/10;
parameter Real omega0=2*pi*F0;
parameter Real k0=alpha^2+(omega0)^2;
parameter Real f0=2*alpha;

Real x_analytical;

equation

x_analytical = exp(-alpha*time) *
               (x0*cos(omega0*time) +
                (alpha*x0/omega0)*sin(omega0*time));

annotation(
  experiment(StartTime=0, StopTime=0.2, Interval=0.001),
  uses(Modelica(version = "4.0.0"))
);

end solution;
```

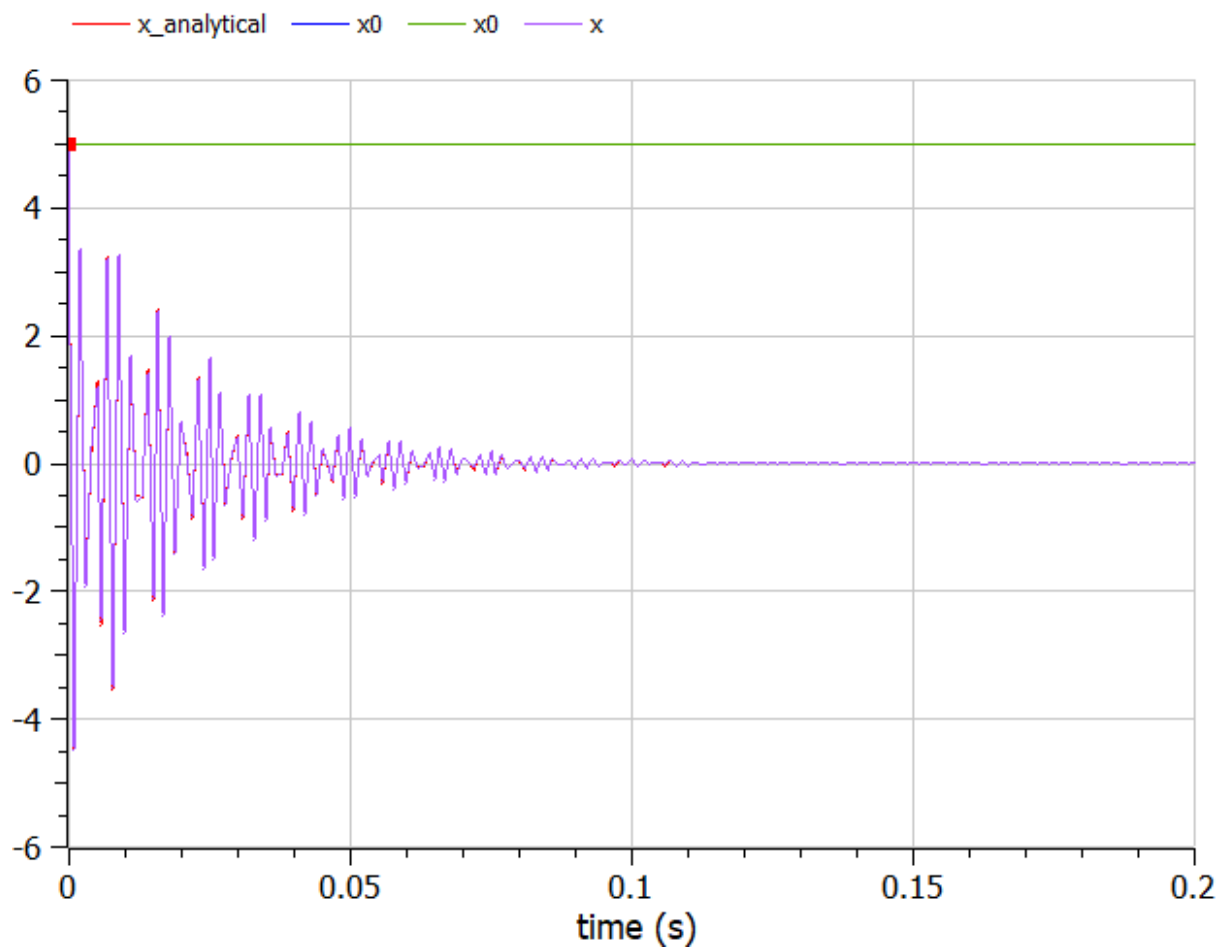


Figure 4: Comparison between the simulation solution and the analytical solution

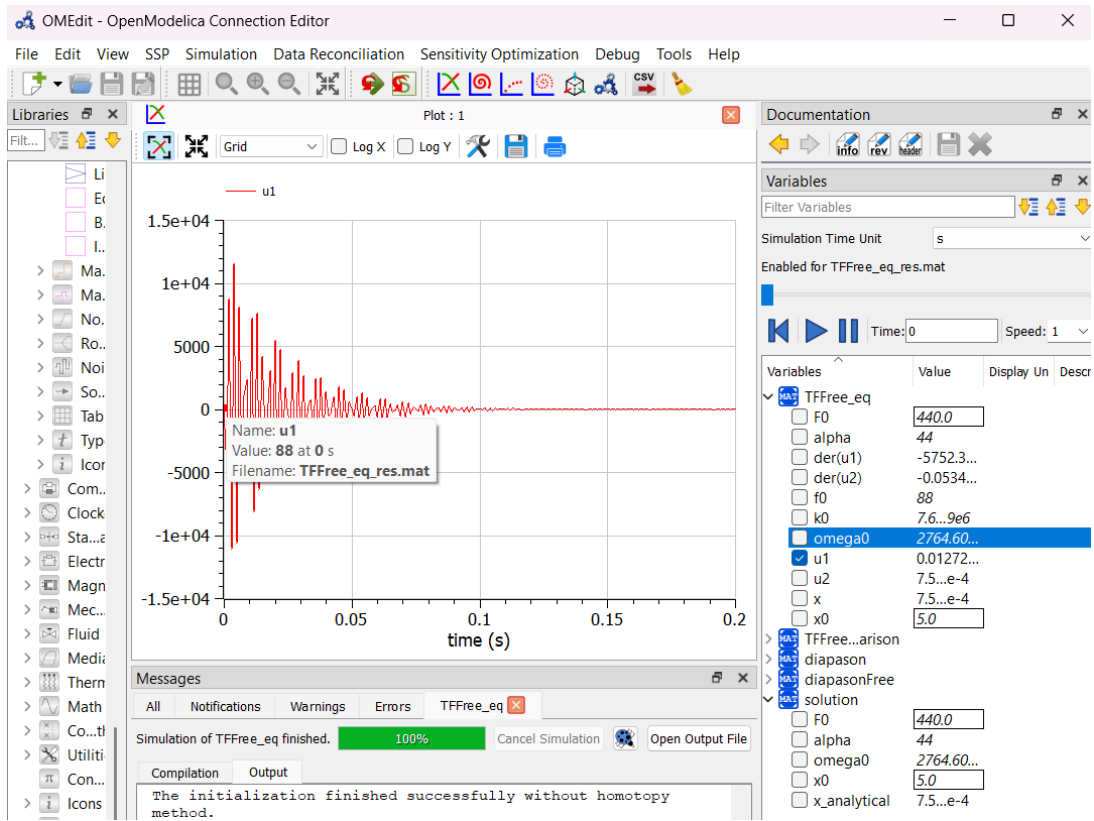


Figure 5: $u_1(0) = 88$, which is not correct ($u_1(0) = 440$, actually).

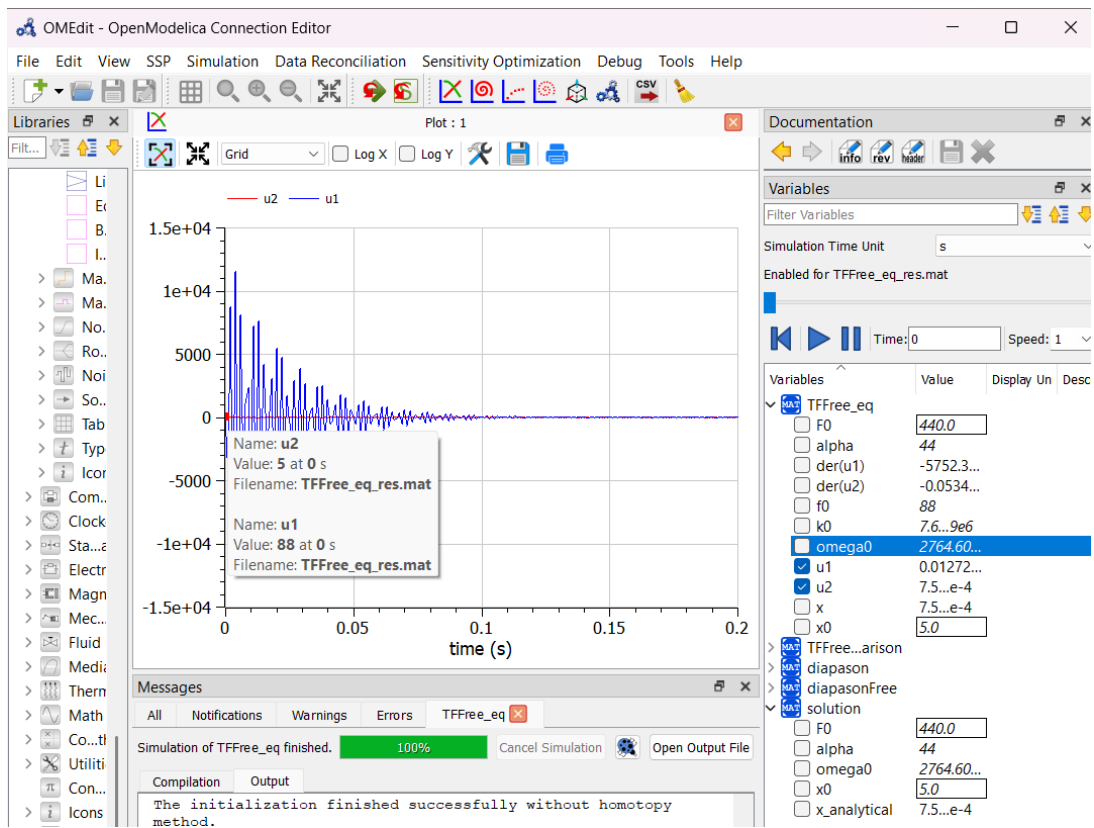


Figure 6: $u_2(0) = 5$, which is correct.

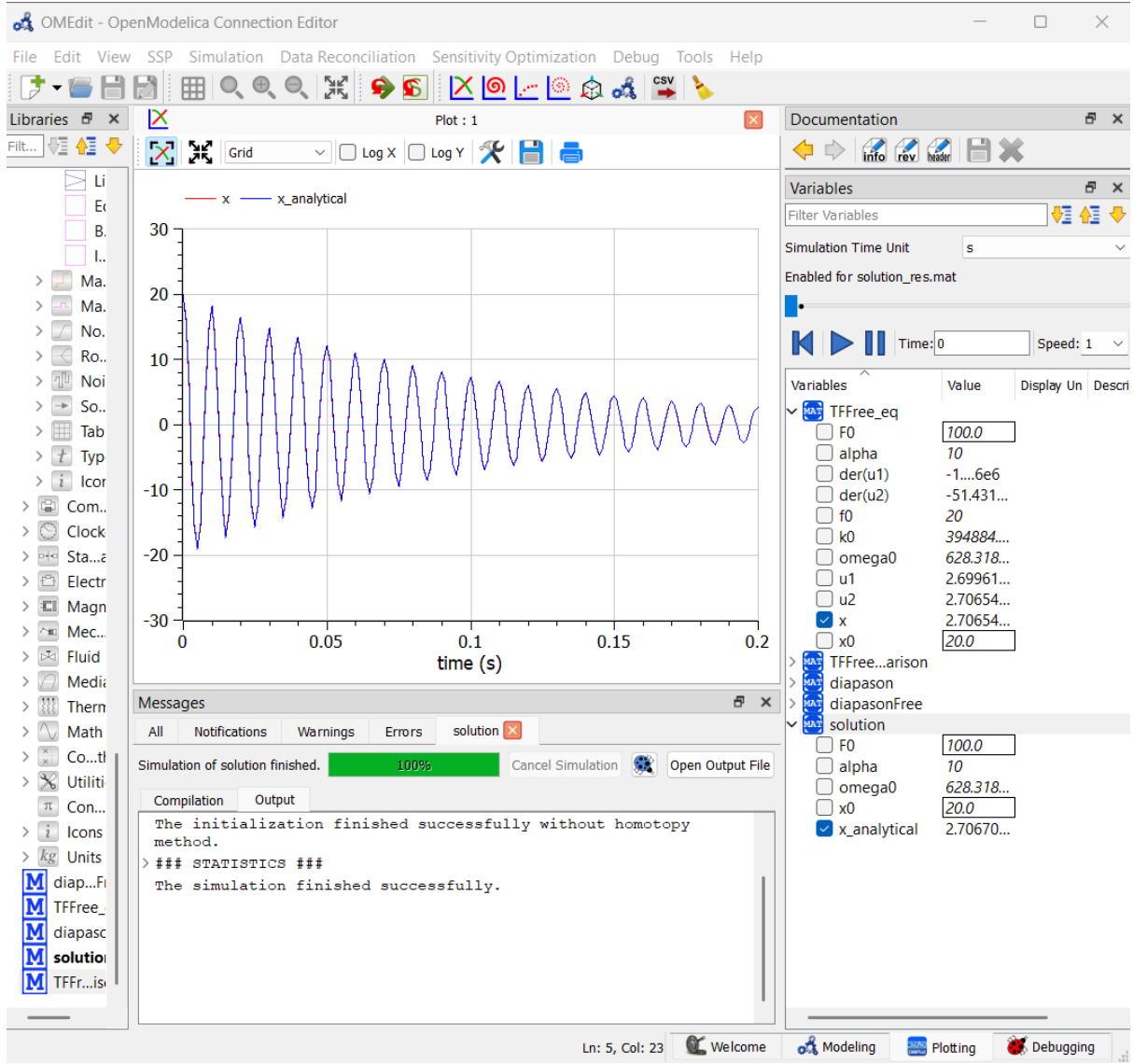


Figure 7: Caption

4. Modify the values of F_0 , x_0 and α . Comment the results obtained.

- Given the values $x_0 = 20$, $F_0 = 100$ and $\alpha = 10$, we obtain the following plot.
- The initial conditions are $x(0) = 20$, $u_2(0) = x_0 = 20$ and $u_1(0) = x'(0) + 2\alpha \times x(0) = 0 + 2 \times 10 \times 20 = 400$.
- The simulation shows that indeed $x(0) = 20$, but $u_2(0) = x_0 = 20$ and $u_1(0) = 20$, so the conditions are not fully correct.
- The characteristic frequency has the value $\omega_0 = 628.318$, which is perfectly equal to $2 \times \pi \times F_0 = 2 \times \pi \times 100 = 628.318$ Hz.
- The damping factor is $\alpha = 10$, the exact same value as $\frac{F_0}{10} = \frac{100}{10} = 10s^{-1}$.

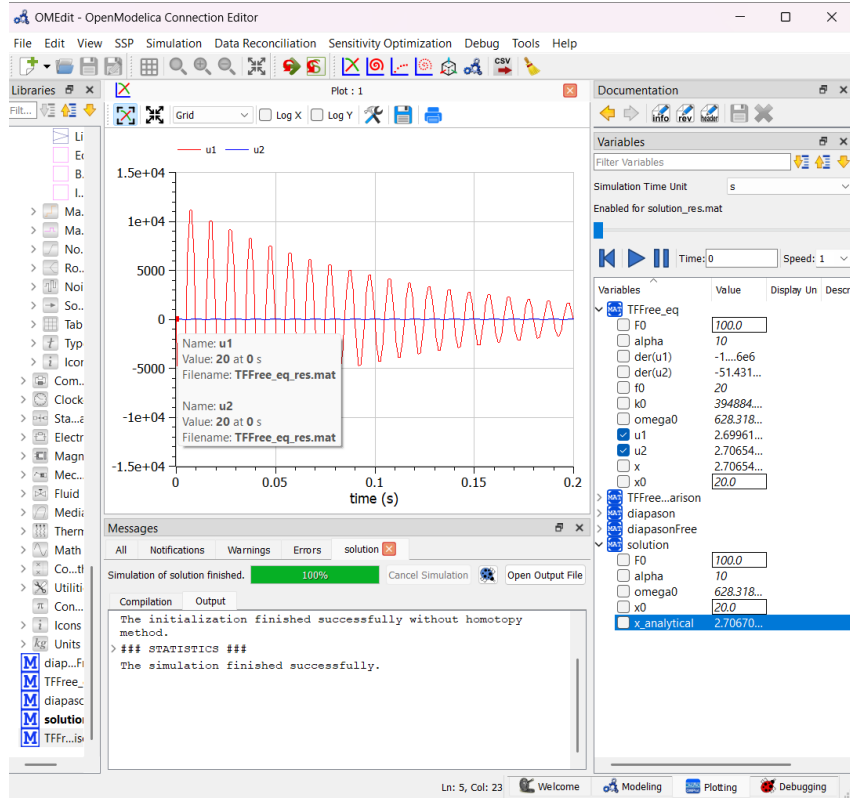


Figure 8: $u_1(0) = 20$, which is not correct ($u_1(0) = 440$, actually).

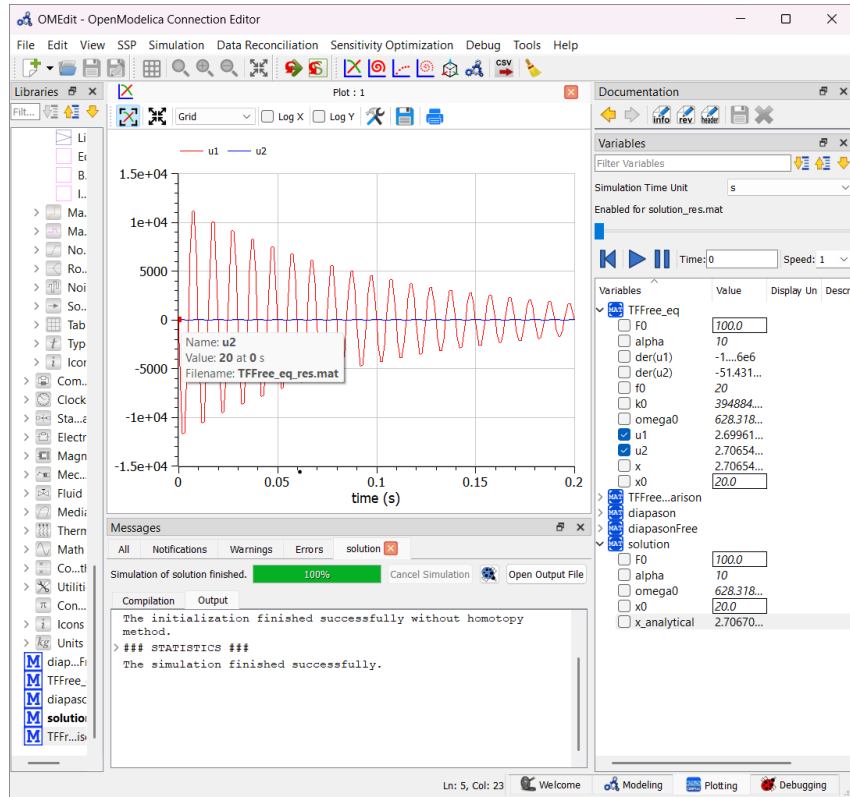


Figure 9: $u_2(0) = 20$, which is correct.

Second situation: $E(t) = A \sin(\omega_E t) \epsilon(t)$

5. Modify the previous Open Modelica program to simulate this situation, with $A = 10^6$. The input signal will get from the sources blocks or with the following instruction:

```
e = A*\sin(2*\pi*Fe*time);

model TFFree_eq

import Modelica.Constants.pi;
parameter Real x0=5;
parameter Real F0=440;
parameter Real alpha=F0/10;
parameter Real omega0=2*pi*F0;
parameter Real k0=alpha^2+(omega0)^2;
parameter Real f0=2*alpha;

parameter Real A = 1e6;
parameter Real Fe = 420;
parameter Real omegaE = 2 * pi * Fe;

Real u1(start =f0);
Real u2(start =x0);
Real x;
Real e;

equation
x=u2;
der(u1)=-k0*x+e;
der(u2)=-f0*x+u1;
e = A * sin(2 * pi * Fe * time);

annotation(
  experiment(
    StartTime=0,
    StopTime=0.15,
    Interval=0.001
  ),
  uses(Modelica(version = "4.0.0"))
);

end TFFree_eq;
```

6. Run a simulation up to 150ms for $F_E = 420\text{Hz}$ and also for $F_E = 440\text{Hz}$ (remember that $\omega_E = 2\pi F_E$). In both cases, observe the output. What is your interpretation?

For $F_E = 420\text{ Hz}$:

- The input frequency (420 Hz) is different from the system's natural frequency (440 Hz).

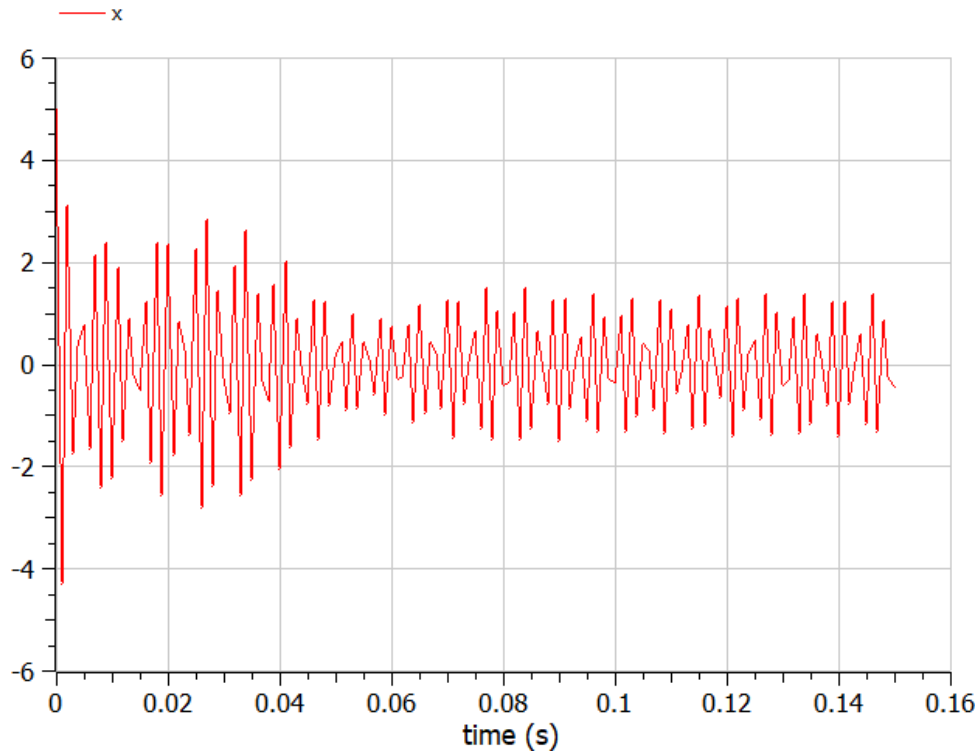


Figure 10: Caption

- The system does not resonate, and the output x shows **damped oscillations** with a relatively small amplitude.
- The response frequency is close to the system's natural frequency (440 Hz), but the amplitude decreases over time due to damping.

For $F_E = 440$ Hz:

- The input frequency (440 Hz) matches the system's natural frequency (440 Hz).
- The system **resonates**, and the output x shows **large oscillations** with a significantly larger amplitude.
- The response is sustained and pronounced, as the system absorbs energy efficiently at its natural frequency.

Interpretation

- When the input frequency matches the system's natural frequency ($F_e = 440$ Hz), resonance occurs, leading to a large amplitude response.
- When the input frequency is different ($F_e = 420$ Hz), the system does not resonate, and the response is smaller and damped.
- This behavior is consistent with the theory of forced oscillations and resonance in dynamical systems.

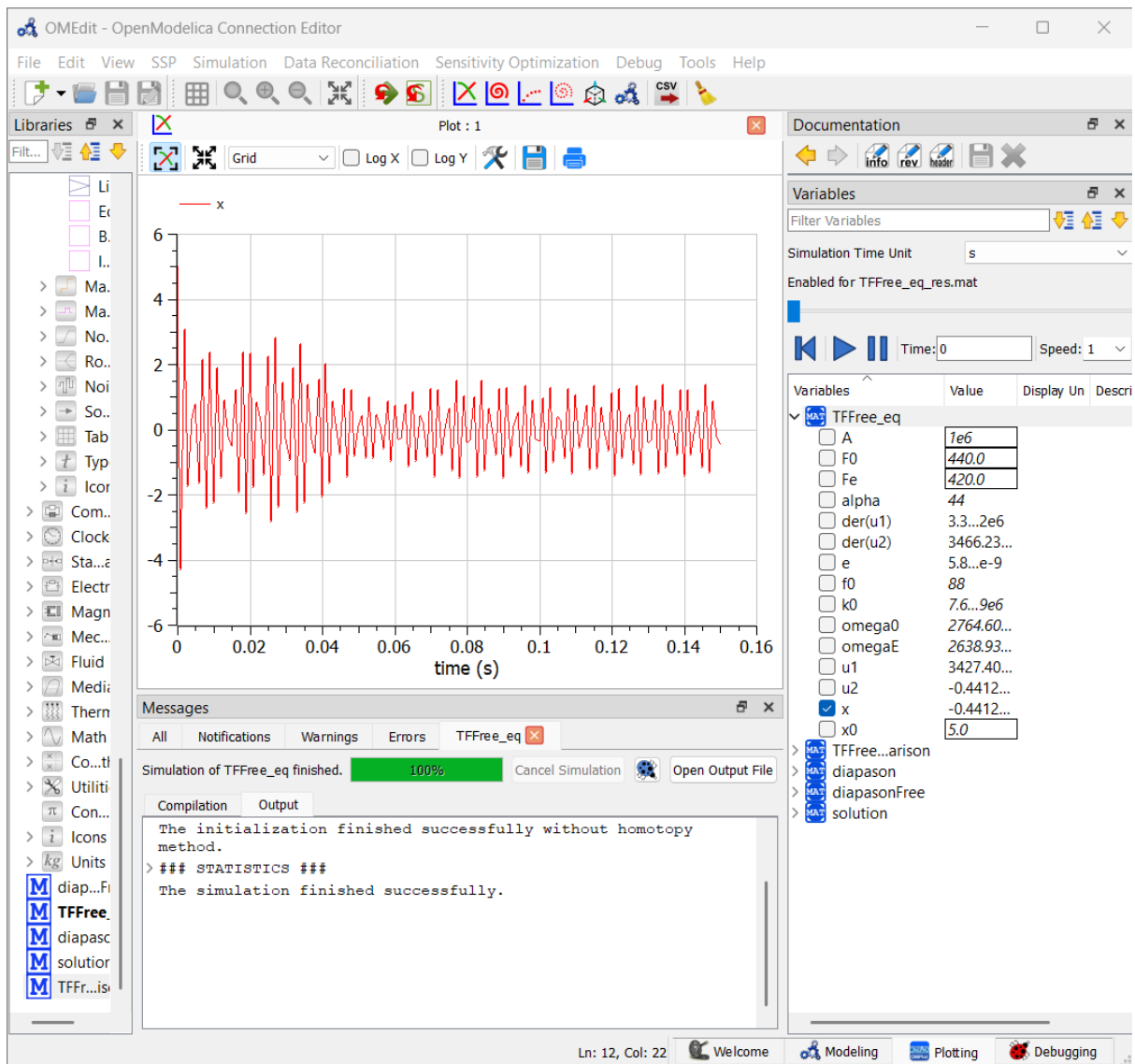


Figure 11: Caption

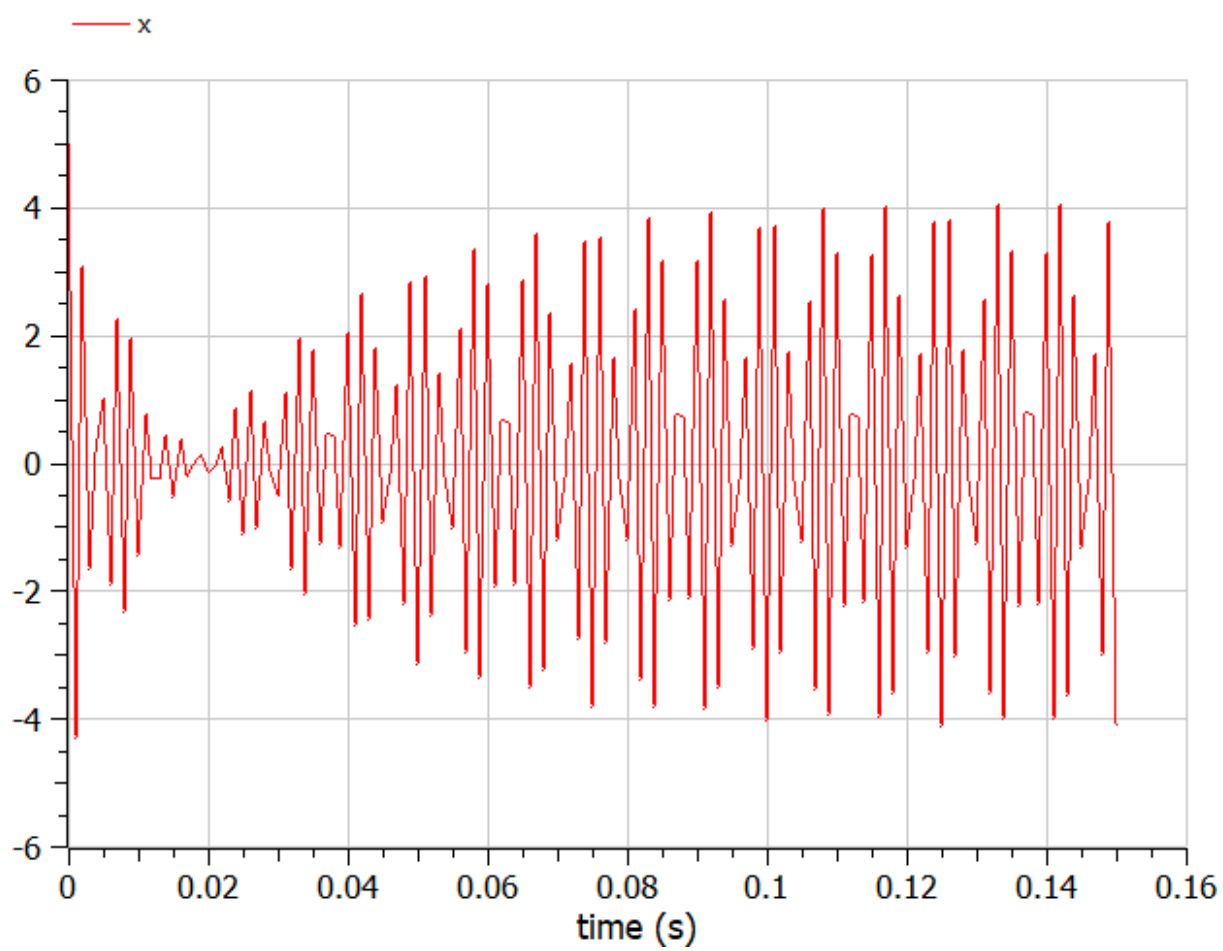


Figure 12: Caption

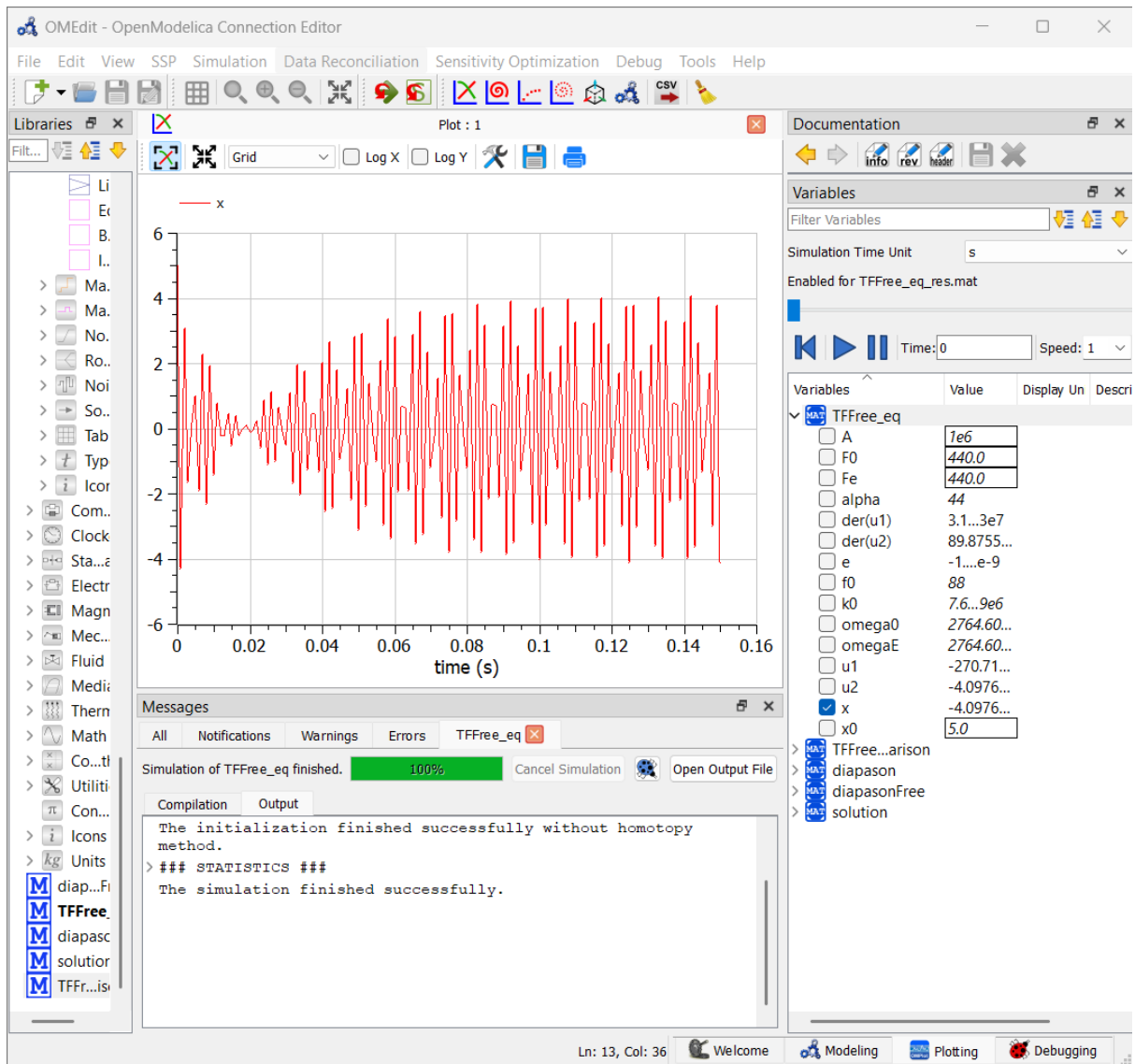


Figure 13: Caption

7. Run a couple of simulations by varying F_E from 420Hz to 460Hz in steps of 5Hz. For each simulation, record the amplitude of the steady-state oscillatory signal (i.e. the zero-state response).

Procedure: use a quadratic detector with a low-pass filter of order 4 and a cut-off frequency (f_c) of 100Hz. The low-pass filter is implemented by the Transfer Function block. In the detector, the signal is firstly squared, then filtered, and then, we take the square root of the signal. To avoid numerical errors when the instantaneous amplitude is very small, the absolute value is computed before the square root.

The transfer function is given by:

$$H(s) = \frac{a^4}{s^4 + 4as^3 + 6a^2s^2 + 4a^3s + a^4},$$

with $a = 2\pi f_c$. So the Transfer Function block will be configured as:

```
Modelica.Blocks.Continuous.TransferFunction transferFunction
(a = {1, 4*a, 6*a^2, 4*a^3, a^4}, b = {a^4})
```

For modeling based on the equations, we need to add:

```
y = x*x; transferFunction.u = y; z = sqrt(abs(2*transferFunction.y));}
```

What do you observe concerning the evolution of the amplitude of the zero-state response estimated at the output of the quadratic detector, versus the frequency F_E ?

```
model TFFree_eq

import Modelica.Constants.pi;
parameter Real x0=5;
parameter Real F0=800;
parameter Real alpha=F0/10;
parameter Real omega0=2*pi*F0;
parameter Real k0=alpha^2+(omega0)^2;
parameter Real f0=2*alpha;

parameter Real A = 1e6;
parameter Real Fe = 420;
parameter Real omegaE = 2 * pi * Fe;

parameter Real fc = 100;
parameter Real a = 2 * pi * fc;
parameter Real num[5] = {a^4, 0, 0, 0, 0};
parameter Real den[5] = {1, 4*a, 6*a^2, 4*a^3, a^4};

Real u1(start =f0);
Real u2(start =x0);
Real x;
Real e;
Real y;
Real y_filtered;
```

```

Real amplitude;

Modelica.Blocks.Continuous.TransferFunction transferFunction(
    a = den,
    b = num
);

equation
x=u2;
der(u1)=-k0*x+e;
der(u2)=-f0*x+u1;
e = A * sin(2 * pi * Fe * time);

y = x^2;
transferFunction.u = y;
y_filtered = transferFunction.y;
amplitude = sqrt(abs(y_filtered));

annotation(
    experiment(
        StartTime=0,
        StopTime=0.15,
        Interval=0.001
    ),
    uses(Modelica(version = "4.0.0"))
);

end TFFree_eq;

```

The values observed buy me after several simulations are:

F_E (Hz)	Amplitude at $t = 0.15s$
420	378591
425	466289
430	564204
435	357654
440	1.06675e+06
445	772975
450	217642
455	227960
460	242989

Interpretation

Resonance Peak

- The amplitude peaks at $F_E = 440$ Hz (1.06675×10^6), indicating **resonance**.
- This is because the input frequency ($F_E = 440$ Hz) matches the system's natural frequency ($F_0 = 440$ Hz).

Amplitude Trend

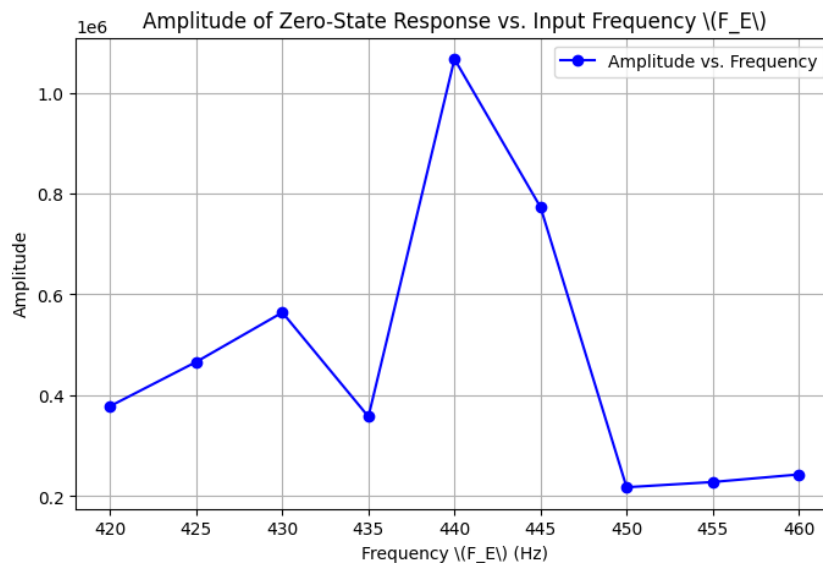
- As F_E increases from 420 Hz to 440 Hz, the amplitude increases, reaching its maximum at 440 Hz.
- As F_E increases beyond 440 Hz, the amplitude decreases.

System Behavior

- The system responds most strongly when the input frequency matches its natural frequency (**resonance**).
- For other frequencies, the system's response is weaker because the input frequency does not match the natural frequency.

The amplitude of the zero-state response evolves as follows:

- The amplitude **increases** as F_E approaches 440 Hz, reaching its maximum value at $F_E = 440$ Hz (1.06675×10^6).
- The amplitude **decreases** as F_E moves away from 440 Hz (both below and above 440 Hz).
- This behavior is due to **resonance** at $F_E = 440$ Hz, where the input frequency matches the system's natural frequency ($F_0 = 440$ Hz), causing the system to absorb energy most efficiently.



8. Change the right parameter of the tuning fork to have a less selective filter. Explain.

To make the tuning fork filter less selective, we need to increase the damping factor α . Here's why and how:

Increase the damping factor α (or the friction coefficient $f_0 = 2\alpha$). For example, if $\alpha = 44 \text{ s}^{-1}$, increase it to $\alpha = 50 \text{ s}^{-1}$ or higher.

- **faster decay of amplitude:** the amplitude of the oscillations $x(t)$ decays more quickly over time.
- **less selective filter:** the system responds to a wider range of frequencies, not just the characteristic frequency ω_0 .

Plot $x(t)$ (displacement) vs. time for two cases:

- **original** α (e.g., $\alpha = 44 \text{ s}^{-1}$): The amplitude decays slowly, and the system is more selective.
- **higher** α (e.g., $\alpha = 50 \text{ s}^{-1}$): The amplitude decays faster, and the system is less selective.

For higher α , the amplitude of $x(t)$ decreases more quickly, and the oscillations die out faster.

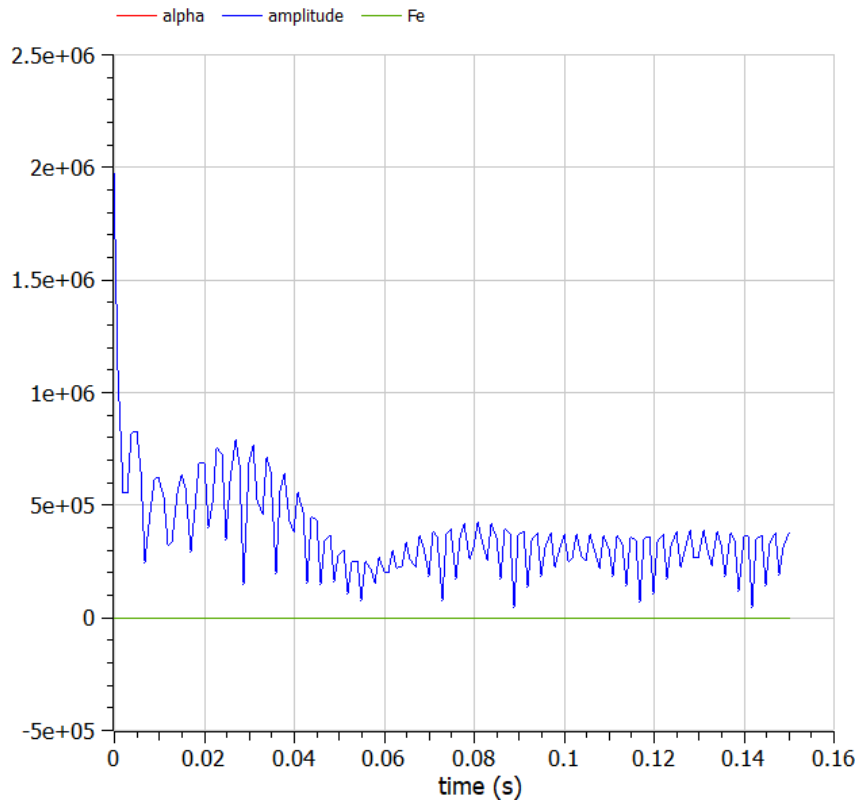


Figure 14: $\alpha = 44$

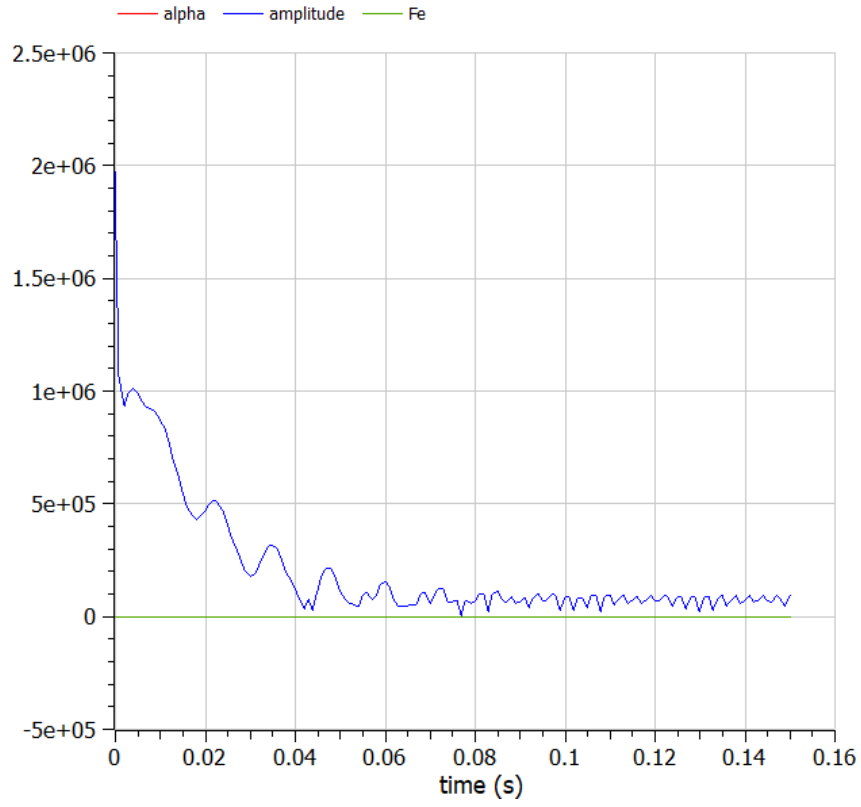


Figure 15: $\alpha = 50$

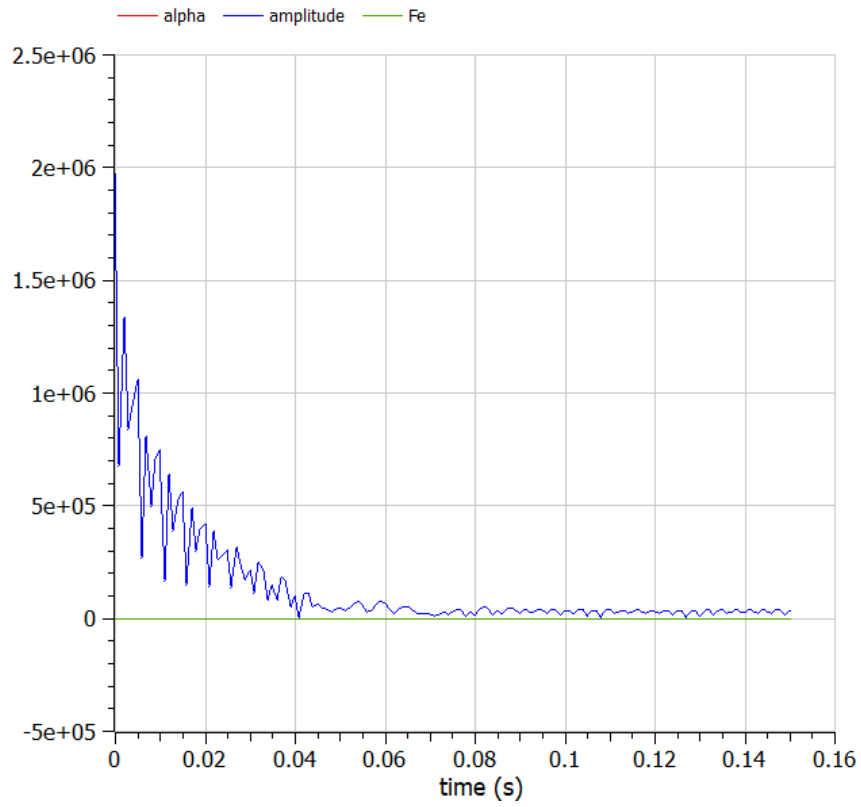


Figure 16: $\alpha = 60$

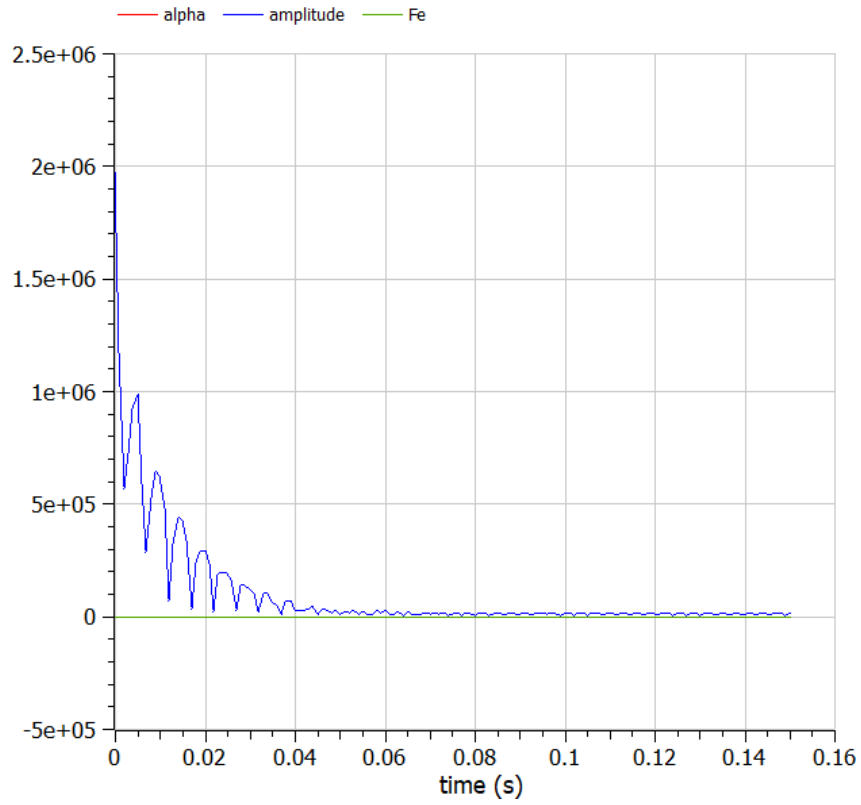


Figure 17: $\alpha = 80$

By increasing the damping factor α , we make the filter less selective. This allows a wider range of frequencies to pass through, and the amplitude of the oscillations decays more quickly.